

Marios Koulakis, PhD

PROFESSIONAL EXPERIENCE

- 2022 – now **Senior Machine Learning Scientist**
DeepHealth / Aidence, Amsterdam – remote from Germany
- 2021 – 2023 **Postdoctoral Researcher, Computer Vision**
Karlsruhe Institute of Technology, CV:HCI Lab, Germany
Full time: 04.2021–06.2022; part time: 07.2022–07.2023.
- 2019 – 2021 **Senior Data Scientist / Data Scientist**
Royal Schiphol Group, Amsterdam Area, Netherlands
Promoted to Senior Data Scientist: 01.2020.
- 2018 **Data Scientist and Engineer**
Nulogy Corporation, Toronto, Canada
- 2016 – 2018 **Software Engineer**
Nulogy Corporation, Toronto, Canada
- 2018 **Big Data Engineering Lecturer**
TranQuant Inc., Toronto, Canada
- 2015 – 2016 **Full-Stack Developer**
ASD Personalinformationssysteme GmbH, Osnabrück, Germany

EDUCATION

- 2010 – 2015 **PhD in Mathematics**
University of Münster, Germany
Dissertation: *Coding into Inner Models at the Level of Strong Cardinals*.
Focus: mathematical logic, set theory, inner model theory, and forcing.
- 2008 – 2010 **MSc in Logic, Algorithms and Computation**
University of Athens, Greece
Thesis: *Large Cardinals and Elementary Embeddings of V*.
- 2002 – 2008 **BSc in Mathematics**
Aristotle University of Thessaloniki, Greece
Focus on pure mathematics.

RESEARCH AND INDUSTRIAL PROJECTS

- 2022 – now **DeepHealth Lung (Chest Suite)**
Co-developed models extending the chest-CT lung-nodule system; contributed to regulatory submissions and mentored junior colleagues.
- 2022 **h-NNE**
Co-conceived the hierarchical projection method; designed its layout algorithm and non-overlap guarantee; implemented most code; 45k downloads.
- 2021 – 2023 **ESKIMO BMBF project**
Led KIT's day-to-day coordination with partners and supported implementation and methodological development.
- 2019 – 2021 **Deep Turnaround**
Proposed the end-to-end video-based direction and co-implemented action-recognition models for turnaround monitoring.



CONTACT

- 📍 Karlsruhe Region, Germany
- 🌐 koulakis.github.io
- 🔗 github.com/koulakis
- 🌐 linkedin.com/in/marioskoulakis

FOCUS AREAS

Data and representation geometry
Manifold and representation learning
DL theory and generalization
Mechanistic interpretability
Robustness and trustworthy scientific AI

LANGUAGE PROFICIENCY

English: Full professional
German: Full professional
Greek: Native
Spanish: Elementary

TEACHING & INTERESTS

Bachata dance instructor
Salsa, bachata, and merengue dancer

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TEACHING

- 2022 – 2023 **Thesis Supervision**
Supervised KIT master and bachelor theses: Wei Zhe and Jonas Bühler.
- 2022 **Deep Learning for Computer Vision II**
Teaching Assistant, KIT, WS 22/23; advanced topics and exercises.
- 2021 – 2022 **Deep Learning for Computer Vision I**
Teaching Assistant, KIT, SS 21 and SS 22; basics and tutorials.
- 2021 **Practical Course Computer Vision for HCI**
Co-organizer, KIT, SS 21; practical computer vision for HCI.
- 2018 **Big Data Engineering Instructor**
Created and taught private big-data engineering workshops in Toronto.

ORGANIZATION AND PUBLIC SPEAKING

- 2025 **Identifying Structure in Data**
Co-organizer and speaker for the CVPR 2025 tutorial on dimensionality reduction, clustering, and related structure-discovery methods.
- 2025 **Dimensionality Reduction and Clustering Tutorial**
Co-organizer and speaker at the internal algorithm meeting of DeepHealth, Amsterdam.
- 2019 **Analysing the Aircraft Turnaround Process Using Deep Learning**
Talk about the Deep Turnaround project at Big Data Expo NL, Amsterdam.

AWARDS

- 2026 **Gold Reviewer Award, ICML 2026**
Awarded to the top 25% reviewers for high-quality peer review.
- 2022 **Teaching Award – Best Practical Course**
KIT Faculty of Informatics award for the CV:HCI practical course in computer vision.
- 2010 – 2015 **Marie Skłodowska-Curie Fellowship**
European fellowship for the MALOA project at the University of Münster.
- 2017 **1st Place, HackOn(Data), Toronto**
Awarded first place in a Toronto data hackathon competition.
- 2002 **2nd Prize, Greek National Mathematical Olympiad**
19th national mathematics olympiad.